

LESSON 139 (0.5) Ground (2.0) Cross-Country Dual (5) Landings

Lesson Objective

The purpose of this lesson is to increase the applicant's night flying and cross-country proficiency.

This night cross-country must be more than 100 NM total distance. This lesson must satisfy the requirements of **14 CFR Part 141 Appendix B 4(b) (1) (ii) (A).**

Pre-Brief

- Review Maneuvers and Procedures to be Performed in Flight
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Student should have completed navigation logs for the planned route with minimal errors
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Aviation Weather Handbook ([FAA-H-8083-28A](#))
- Advisory Circulars.
 - Pilot's Guide to a Preflight Briefing (AC91-92)
- Aeronautical Information Manual
 - Chapter 1: Air Navigation
 - Chapter 3: Airspace
 - Chapter 4: Air Traffic Control
 - Section 3: Airport Operations
 - Chapter 6: Emergency Procedures
 - Chapter 7: Safety of Flight
- Pilot's Information Manual
 - Section 4: Normal Procedures
 - Section 5: Performance

Instructor Guidance

- Thoroughly review the student's navigation logs before the flight to ensure they are completed correctly and contain no errors. Do not depart until the log is accurate — a navigation log with mistakes on a 100+ NM night cross-country is a safety risk, not just an administrative one.
- Conduct an in-depth briefing covering all components of 14 CFR §91.103 preflight action requirements. The student must be able to explain how the existing weather, NOTAMs, and conditions will affect the flight before being cleared to go.
- File flight plans with FSS for both directions of flight before departing. Ensure the student participates in this process and understands how to open and close flight plans both in the air and on the ground. The flight plan must be closed upon arrival — an open flight plan after 30 minutes will trigger search and rescue.

- Ensure the student is comfortable making calls to both FSS and ATC for flight following before departing. If there are any deficiencies in radio phraseology, address them on the ground — night operations increase cockpit workload and leave less bandwidth for radio difficulties.
- Have a thorough discussion on diversion procedures during the pre-brief. The student will face a simulated diversion scenario on the return leg — they must know how to locate their position, select an alternate airport, fill out the diversion log, and amend the flight plan before the scenario is introduced in the air.
- Emphasize the importance of taxiing slower at night. Depth perception and peripheral visual cues are significantly reduced in darkness — a student who taxis at their normal daytime speed at an unfamiliar airport is taking an unnecessary risk.
- Review night operations procedures per the FOM before departing and ensure the student is familiar with all Florida Tech -specific night requirements and restrictions.

Lesson Content

Checklist Procedures

- Have the student run through all checklists.
 - Ensure PFD is correctly configured for departure with the correct route of flight and altitude.
- Incorporate flows for memory items.
 - Cruise Checklist must be done when leveled off in flight as a memory item.
- Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student can proficiently use COMS and NAV units.
 - Student should be able to correctly tune and ID a VOR.
 - Have the student put the appropriate approach frequency in to pick up flight following.
- Confirm the student has correctly input the flight plan for the route planned and verify it is shown correctly on the map page.

Engine Starting

- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls at towered and non-towered airports.
- The student should be able to conduct radio calls in the practice area.
- Student should be able to open and close a flight plan with FSS:
 - Opening the flight plan:

- Pilot: "St-Petersburg radio, N612FT, transmitting and receiving on 122.2".
- FSS: "N612FT, St. Petersburg radio, go ahead".
- Pilot: "St-Petersburg radio, N612FT would like to open our flight plan from KMLB to KOBE".
- FSS: "N612FT, St. Petersburg radio, flight plan is activated".
- Closing the flight plan:
 - Pilot: "St-Petersburg radio, N612FT, transmitting and receiving on 122.2"
 - FSS: "N612FT, St. Petersburg radio, go ahead".
 - Pilot: "St-Petersburg radio, N612FT would like to close our flight plan from KMLB to KOBE".
 - FSS: "N612FT, St. Petersburg radio, flight plan is terminated".
- Student should be able to pick up VFR flight following:
 - Picking up:
 - Pilot: "Orlando approach, TCF612 6 miles South-West of Melbourne, requesting VFR flight following to KOBE, Altitude 3000ft, Type Papa 28 Alpha"
 - ATC: "TCF612, Orlando Approach, squawk 1234"
 - Pilot: "Squawk 1234, TCF612"
 - ATC: "TCF612, Orlando Approach, radar contact 6 miles South-West of Melbourne, Altimeter 30.01"
 - Cancelling:
 - Pilot: "Orlando approach, TCF612, requesting to cancel VFR Flight following"
 - ATC: "TCF612, Orlando approach, radar services terminated, squawk VFR, frequency change approved"
 - Pilot: "TCF612, radar services terminated, squawk VFR, frequency change approved"

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
 - Flight following can assist with traffic alerts but PIC is still responsible to see and avoid.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Taxiing

- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance
- Ensure the correct use of wind correction throughout taxi.
- Verify the student has airport diagrams for new airports and ensure they brief all taxi instructions.
 - Briefing taxi routes is always required but should be emphasized at unfamiliar airports.

Normal and Crosswind Takeoff

- Takeoff procedures at night are the same as during the day.
 - If able, avoid rolling takeoffs at night.
 - Intersection departures are NOT ALLOWED.
- Areas of emphasis:
 - Clear the area thoroughly. It is harder to identify a clear runway and approach and

- departure at night.
- Ensure the magnetic compass and HSI match with the intended runway centerline.
- Verify line up check is complete.
 - Check that centerline is maintained using the lights to verify when able.
 - Confirm that runway edge lights are correctly spaced at both sides from centerline.
- Perception of runway width, airplane speed, and flight attitude will vary at night.
 - Check flight instruments frequently.
- The ensure airplane is climbing, check the airspeed indicator, VSI, and altimeter;
 - Darkness makes it hard to judge rate of climb.
- Ensure turns are made at safe altitudes.

Pilotage and Dead Reckoning:

- Verify the student uses both pilotage and dead reckoning throughout the flight and corrects heading to track points as necessary.
- Pilotage: navigation by reference to landmarks or checkpoints.
 - A method of navigation that can be used on any course with adequate checkpoints but is more commonly used with dead reckoning and VHF radio navigation.
 - The checkpoint used should be prominent features common to the area of flight.
 - Roads, rivers, railroad tracks, lakes, and power lines.
 - If possible, select features that will make useful boundaries on each side of the course.
 - Keep from drifting too far off course by referring to and not crossing selected points.
 - Never place complete reliance on any single checkpoint, choose ample checkpoints.
 - If the checkpoint is not in sight, do not continue blindly.
- Dead reckoning: navigation solely by computation based on time, airspeed, distance, and direction.
 - The products derived from these, when adjusted by wind speed and velocity, are heading and ground speed (GS).
 - The predicted heading will guide the airplane along the intended path and the GS will establish the time to arrive at each checkpoint and destination.

Navigation Systems and Radar Services

- Navigation Systems
 - Verify all GPS databases are up to date on the G430 or G650.
 - Student should be able to correctly input all required frequencies and navigate correctly using the nearest page.
 - Student should be able to input flight plan and verify the CDI is on the correct mode to display either GPS or VOR information on the G5 or Avidyne.
 - Student should be able to correctly tune and ID a VOR.
 - For the G5:
 - Once tune and ID is complete, press the CDI button on the G430 or G650 to switch it from GPS to VLOC.

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- The G5 HSI should then show the green VOR needle.
- To adjust the course, press the right knob on G5, scroll to the right to CRS, and press the right knob. This allows the pilot to twist to the desired course.
- For the Avidyne:
 - Once tune and ID is complete, press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired VLOC.
 - To adjust the course, use the left knob on the PFD to adjust the needle to the desired course. To center it, simply press the left knob.
- Ensure the student knows the difference between an outbound and inbound indication.
 - Outbound: arrows point away from each other.
 - Inbound: arrows point the same way.
 - When given an outbound instruction like "TCF609 intercept and track the 360-radial outbound", the course should be set to 360.
 - When given an inbound instruction like "TCF609 intercept and track the 360-radial inbound", the tail should be set to 360 OR take the reciprocal of 360 and set that number to the course.
- For the purpose of this lesson, the student should be able to:
 - Figure out the current radial.
 - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.
- Radar Services
 - Understand primary radar (range and azimuth only) and secondary radar (transponder-based — Mode A/C/S).
 - Transponder codes:
 - 1200 – VFR, 7500 – Hijack, 7600 – Radio Failure, 7700 – Emergency.
 - When working with ATC radar:
 - Listen for assigned discrete transponder code.
 - Respond to radar vectors using standard-rate turns and precise altitude control.
 - Read back headings and altitudes to ensure clarity.
 - If transitioned to a different ATC sector:
 - ATC will provide a frequency to change to. Check in by transmitting "Palm Beach Approach, TCF612, 3000ft"
 - If given a heading to follow, state the heading you are on.
 - If unable to hear ATIS/AWOS when attempting to pick it up or only operating using one radio, request a frequency change with ATC; ATC may give you a time frame to check back in with them.

Navigation Log

- Departure
 - Ensure flight plan, radios, and NAVaids are correctly configured.

- Complete Runway Entry Check:
 - Note TIME OFF on navlog and fill out ETA of the first point.
- Follow ATC instruction to proceed on course.
- Cruise
 - When leveled off at correct altitude, complete Cruise Check
 - Ensure the student knows the proper leaning procedures for Best Power vs. Best Economy
 - Make sure student is switching tanks when required
 - Evaluate CRM and division of attention
 - Ensure traffic scanning when filling out navlog
 - Pilotage and Dead Reckoning
 - Student should verbally verify checkpoints
 - Ensure course is intercepted, bug and follow planned heading
 - Record actual Ground speed
 - ETAs, ATAs, ETEs, ATEs all recorded on navlog at each checkpoint
 - If incorrect or inaccurate, revise using E6B
 - Fill out fuel burn
 - Radio Communication
 - Opening Flight Plan
 - Picking up Flight Following
 - Set required frequencies when in between points
- Arrival
 - Pick up ATIS/AWOS of destination airport when able
 - Start descent at calculated TOD
 - Do Descent and Arrival checklist prior to descent.
 - Thoroughly brief arrival for pattern entry, expected runway, and runway lengths.
 - Monitor frequency to determine runways in use.
 - If non-towered, make radio call 10 miles from airport to report position and intentions.
 - Radio Communication
 - Announce intent to descend to ATC.
 - Contact Flight Service to close flight plan.
 - Complete navlog when safe to do so.

Diversion

- On the return leg, simulate a scenario where a diversion is necessary.
- Diversion Log
 - Have student note time of diversion.
 - Ensure student can locate present position using pilotage and dead reckoning and make note on the sectional.
 - Choose an alternate airport to divert to by selecting the closest/most appropriate airport shown on the sectional or 'Nearest Page' on the G430 or G650.
 - The student must divert immediately toward the diversion airport using shortcuts/rule

of thumb calculations if it is an emergency. If time permits, the student may circle until all information is obtained.

- Student must correctly fill out the diversion log based on the scenario given:
 - Fill out heading, distance, time enroute, fuel burn, and checkpoints.
 - Amend flight plan as necessary and update ATC if on flight following.
 - Ensure the student uses the chart supplement to get frequencies and runway information for the diversion airport.
- Stress the importance of positive aircraft control.
 - Ensure the student shows situational awareness and continues to scan for traffic.

Lost Procedures

- When departing, on the second leg, direct the student to an unfamiliar area.
- Take away the student's sectional chart, GPS course, and radio navigation.
- Present the lost scenario.
- Give back navigation equipment.
- Ensure student says and does the correct lost procedures, (7C's).

Flight Service Station Flight Plan

- During pre-brief, review the student's flight plan on the back of their navlog; correct any mistakes.
- Have the student file the flight plan for both directions of flight via 1800wxbrief.com, by calling 1-800-WXBRIEF, through Foreflight, or through SkyVector. Monitor to ensure accuracy.
- Suggest that the student signs up for EasyActivate/EasyClose so they may open and close flight plans through their phone or email.
- To open or close the flight plan in their air, tune to the correct flight service frequency and refer to the section on **Radio Communication Procedures and ATC Phraseology** for what to say to the FSS.
 - If already on Flight Following, request a frequency change with ATC to open or close plan.
 - Flight plan can be closed on ground at destination airport by calling or using EasyClose.
- DO NOT FORGET TO CLOSE YOUR FLIGHT PLAN!
 - Flight service will attempt to contact you if the flight plan is still open.
 - After 30 minutes of no response, Flight Service may send search and rescue to attempt to locate your aircraft.
- Flight Service can also provide in-flight weather briefings if unforecasted weather appears along your route of flight.

Systems and Equipment Malfunctions (simulated electrical failure)

- Simulate an electrical failure by dimming the PFD.
- Ensure the student can run through the proper checklists and verify them.
- Ensure the student is reducing the electrical load as necessary.
- Question the student of their decision making.
 - Is this a land "as soon as practical" or land "as soon as possible" emergency?

Go-Arounds or Rejected Landings

- Have your student practice go-around procedures.
- Encourage your student to call for a go-around themselves rather than waiting for you to call it.
- If you do call for a go-around, the student should immediately execute the procedure rather than discussing it first.

Normal and Crosswind Landings

- Student should conduct a normal/crosswind landing.
- Ensure runway heading is verified and final check is completed.
 - Verify the magnetic compass and HSI match with the intended runway centerline.
 - Perception of runway width, airplane speed, and flight attitude will vary at night.
 - Check flight instruments frequently.
 - Use approach lighting systems to maintain desired glide path.
 - Check chart supplement prior to flight to see available lighting.
 - Verify lights are working/usable via NOTAMs.
 - Judgement of height, speed, and sink rate may be impaired.
 - There is a tendency to round out too high.
 - Start the round out when the light starts to reflect off the tire marks on the runway.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Completion Standards

This lesson is complete when the applicant can:

1. Conduct a cross-country to a minimum of three points (one must be at a towered airport other than the original point of departure) with one segment of the flight consisting of a straight-line distance of more than 50 NM.
2. Before the flight:
 - a. Complete a navigation log with no errors.
 - b. Complete the climb, cruise, and descent performance calculations for the planned flight.
 - c. Obtain a weather briefing and explain how the existing weather, NOTAMs and conditions will affect the flight.
3. Without instructor guidance:
 - a. Conduct all normal and emergency checklist procedures.
 - b. Intercept the planned course by using an on-course check point.
 - c. Have predetermined ETAs accurate to ± 5 minutes with instructor guidance.
 - d. Calculate groundspeed ± 10 knots based on actual time between check points.
 - e. Maintains altitude ± 200 ft from desired altitude and ± 15 degrees from desired heading.
 - f. Use radio navigation, pilotage, and dead reckoning procedures.
 - g. Maintain directional control during takeoffs and landings.
 - h. Maintain a stabilized approach.

- i. Conduct short field landings and touch down at or within 400 ft beyond a specified point
- j. Conduct soft field landings and touch down softly with no drift.